

❖ Proof that all relations are in Boyce-Codd Normal Form

1. “Content” Relation:

- **Attributes:**

Content { Content_ID, Title, Content_type, Release_date, Age_rating, IMDB_rating, Production_ID }

- **Minimal set of Functional dependencies:**

Content_ID $\rightarrow$ Title

Content_ID $\rightarrow$ Content_type

Content_ID $\rightarrow$ Release_date

Content_ID $\rightarrow$ Age_rating

Content_ID $\rightarrow$ IMDB_rating

Content_ID $\rightarrow$ Production_ID

let $X = \text{Content_ID}$,

$X^+ = \{ \text{Content_ID}, \text{Title}, \text{Content_type}, \text{Release_date}, \text{Age_rating}, \text{IMDB_rating}, \text{Production_ID} \}$

Thus, **The primary key is Content_ID.**

The left side of all the FDs in the minimal set of FDs for the relation ‘Content’ is Content_ID, which is the primary key of this relation,
So “Content” is in BCNF.

2. “Content_Subtitle” Relation:

- **Attributes:**

Content_Subtitle {Content_ID, Subtitle}

Here **Primary key = {Content_ID, Subtitle}**

According to theorem, all attribute primary key relation is always in BCNF.

Hence **“Content_Subtitle” is in BCNF.**

3. "Content_Genre" Relation:

- **Attributes:**

Content_Genre {Content_ID, Genre}

Here **Primary key** = {Content_ID, Genre}

According to theorem, all attribute primary key relation is always in BCNF.

Hence "**Content_Genre**" is in BCNF.

4. "Content_Audio" Relation:

- **Attributes:**

Content_Audio {Content_ID, Audio}

Here **Primary key** = {Content_ID, Audio}

According to theorem, all attribute primary key relation is always in BCNF.

Hence "**Content_Audio**" is in BCNF.

5. "Streams" Relation:

- **Attributes:**

Streams {Stream_ID, Stream_date, Stream_Start_time, Duration, Content_ID, User_ID}

- **Minimal set of Functional dependencies:**

Stream_ID $\rightarrow$ Stream_date

Stream_ID $\rightarrow$ Stream_Start_time

Stream_ID $\rightarrow$ Duration

Stream_ID $\rightarrow$ Content_ID

Stream_ID $\rightarrow$ User_ID

Take X = Stream_ID,

$X^+ = \{\text{Stream_ID, Stream_date, Stream_Start_time, Duration, Content_ID, User_ID}\}$

That's why, **The primary key is Stream_ID.**

The left side of all the FDs in the minimal set of FDs for the relation 'Streams' is Stream_ID, which is the primary key of this relation,
So "Streams" is in BCNF.

6. "Watchlist" Relation:

- **Attributes:**

Watchlist {User_ID, Watchlist}

Here **Primary key = {User_ID, Watchlist}**

According to theorem, all attribute primary key relation is always in BCNF.

Hence **"Watchlist" is in BCNF.**

7. "Statistics" Relation:

- **Attributes:**

Statistics {Content_ID, Total_views, Revenue, Budget}

- **Minimal set of Functional dependencies:**

Content_ID $\rightarrow$ Total_views

Content_ID $\rightarrow$ Revenue

Content_ID $\rightarrow$ Budget

Take X = Content_ID,

$X^+ = \{\text{Content_ID, Total_views, Revenue, Budget}\}$

That's why, **The primary key is Content_ID.**

The left side of all the FDs in the minimal set of FDs for the relation 'Statistics' is Content_ID, which is the primary key of this relation,
So "Statistics" is in BCNF.

8. “Related_Shows” Relation:

- **Attributes:**
Related_Shows {Show_ID, Content_ID, Similarity}
- **Minimal set of Functional dependencies:**
 $\{ \text{Show_ID}, \text{Content_ID} \} \rightarrow \text{Similarity}$

Take $X = \{ \text{Show_ID}, \text{Content_ID} \}$

$X^+ = \{ \text{Show_ID}, \text{Content_ID}, \text{Similarity} \}$

That’s why, **The key is {Show_ID, Content_ID}.**

The left side of all the FDs in the minimal set of FDs for the relation ‘Related_Shows’ is {Show_ID, Content_ID}, which is the key of this relation,

So “Related_Shows” is in BCNF.

9. “Production” Relation:

- **Attributes:**
Production {Production_ID, Name, PAN_No, Location}
- **Minimal set of Functional dependencies:**
 $\text{Production_ID} \rightarrow \text{Name}$
 $\text{Production_ID} \rightarrow \text{PAN_No}$
 $\text{Production_ID} \rightarrow \text{Location}$

Take $X = \text{Production_ID}$

$X^+ = \{ \text{Production_ID}, \text{Name}, \text{PAN_No}, \text{Location} \}$

That’s why, **The primary key is Production_ID.**

The left side of all the FDs in the minimal set of FDs for the relation ‘Production’ is Production_ID, which is the primary key of this relation, **So “Production” is in BCNF.**

10. “Advertisement” Relation:

- **Attributes:**

Advertisement {Advertise_ID, Product_Name, Company_Name, AD_cost, Views, Revenue}

- **Minimal set of Functional dependencies:**

Advertise_ID $\rightarrow$ Product_name

Advertise_ID $\rightarrow$ Company_name

Advertise_ID $\rightarrow$ AD_cost

Advertise_ID $\rightarrow$ Views

Advertise_ID $\rightarrow$ Revenue

Take $X = \text{Advertise_ID}$

$X^+ = \{\text{Advertise_ID}, \text{Production_Name}, \text{Company_Name}, \text{AD_cost}, \text{Views}, \text{Revenue}\}$

That's why, **The primary key is Advertise_ID.**

The left side of all the FDs in the minimal set of FDs for the relation 'Advertisement' is Advertise_ID, which is the primary key of this relation, **So “Advertisement” is in BCNF.**

11. “Advertisement_Genre” Relation:

- **Attributes:**

Advertisement_Genre {Advertise_ID, Genre}

Here **Primary key = {Advertise_ID, Genre}**

According to theorem, all attribute primary key relation is always in BCNF.

Hence **“Advertisement_Genre” is in BCNF.**

12. “Actor” Relation:

- **Attributes:**

Actor {Actor_ID, Name, DOB, Gender, Nationality}

- **Minimal set of Functional dependencies:**

Actor_ID $\rightarrow$ Name

Actor_ID $\rightarrow$ DOB

Actor_ID $\rightarrow$ Gender

Actor_ID $\rightarrow$ Nationality

Take X = Actor_ID

$X^+ = \{\text{Actor_ID, Name, DOB, Gender, Nationality}\}$

That’s why, **The primary key is Actor_ID.**

The left side of all the FDs in the minimal set of FDs for the relation ‘Actor’ is Actor_ID, which is the primary key of this relation,

So “Actor” is in BCNF.

13. “Tv_Series” Relation:

- **Attributes:**

TV_Series {Content_ID, First_Air_Date, Total_seasons, End_Date}

- **Minimal set of Functional dependencies:**

Content_ID $\rightarrow$ First_Air_Date

Content_ID $\rightarrow$ Total_seasons

Content_ID $\rightarrow$ End_Date

Take X = Content_ID

$X^+ = \{\text{Content_ID, First_Air_Date, Total_seasons, End_Date}\}$

That’s why, **The primary key is Content_ID.**

The left side of all the FDs in the minimal set of FDs for the relation 'TV_Series' is Content_ID, which is the primary key of this relation,
So "TV_Series" is in BCNF.

14. "Episodes" Relation:

- **Attributes:**

Episodes {Episode_No, Season_No, Title, Release_Date, Discription, Rating, Duration, Content_ID}

- **Minimal set of Functional dependencies:**

{ Episode_No, Season_No, Content_ID } $\rightarrow$ Release_Date

{ Episode_No, Season_No, Content_ID } $\rightarrow$ Discription

{ Episode_No, Season_No, Content_ID } $\rightarrow$ Rating

{ Episode_No, Season_No, Content_ID } $\rightarrow$ Duration

{ Episode_No, Season_No, Content_ID } $\rightarrow$ Title

Take $X = \{ \text{Episode_No, Season_No, Content_ID} \}$

$X^+ = \{ \text{Episode_No, Season_No, Title, Release_Date, Discription, Rating, Duration, Content_ID} \}$

That's why, **The key is = { Episode_No, Season_No, Content_ID }.**

The left side of all the FDs in the minimal set of FDs for the relation 'Episodes' is = { Episode_No, Season_No, Content_ID }, which is the key of this relation,

So "Episodes" is in BCNF.

15. "Cast" Relation:

- **Attributes:**

Cast {Role, Actor_ID, Content_ID}

Here **Primary key = { Role, Actor_ID, Content_ID }**

According to theorem, all attribute primary key relation is always in BCNF.

Hence "Cast" is in BCNF.

16. "Shows" Relation:

- **Attributes:**

Shows {Content_ID, Duration}

- **Minimal set of Functional dependencies:**

Content_ID $\rightarrow$ Duration

Take X = Content_ID

$X^+ = \{\text{Content_ID, Duration}\}$

That's why, **The primary key is Content_ID.**

The left side of all the FDs in the minimal set of FDs for the relation 'Shows' is Content_ID, which is the primary key of this relation,
So "Shows" is in BCNF.

17. "Movies" Relation:

- **Attributes:**

Movies {Content_ID, Duration}

- **Minimal set of Functional dependencies:**

Content_ID $\rightarrow$ Duration

Take X = Content_ID

$X^+ = \{\text{Content_ID, Duration}\}$

That's why, **The primary key is Content_ID.**

The left side of all the FDs in the minimal set of FDs for the relation 'Movies' is Content_ID, which is the primary key of this relation,
So "Movies" is in BCNF.

18. "Ott_User" Relation:

- **Attributes:**

Ott_User {User_ID, Name, Gender, DOB, Email, Contact_No, Street, City, Start_date, Subscription_type, Duration}

- **Minimal set of Functional dependencies:**

User_ID $\rightarrow$ Name

User_ID $\rightarrow$ Gender

User_ID $\rightarrow$ DOB

User_ID $\rightarrow$ Email

User_ID $\rightarrow$ Contact_No

User_ID $\rightarrow$ Street

User_ID $\rightarrow$ City

User_ID $\rightarrow$ Start_date

User_ID $\rightarrow$ Subscription_type

User_ID $\rightarrow$ Duration

Take X = User_ID

$X^+ = \{ \text{User_ID, Name, Gender, DOB, Email, Contact_No, Street, City, Start_date, Subscription_type, Duration} \}$

That's why, **The primary key is User_ID.**

The left side of all the FDs in the minimal set of FDs for the relation 'Ott_User' is User_ID, which is the primary key of this relation, so **"Ott_User" is in BCNF.**

19. "Starred_in" Relation:

- **Attributes:**

Starred_in {Actor_ID, Content_ID}

Here **Primary key** = { Actor_ID, Content_ID }

According to theorem, all attribute primary key relation is always in BCNF.

So "Starred_in" is in BCNF.

20. “Displayed_in” Relation:

- **Attributes:**

Displayed_in {Advertise_ID,Content_ID}

Here **Primary key** = { Advertise_ID,Content_ID }

According to theorem, all attribute primary key relation is always in BCNF.

So “Displayed_in” is in BCNF.

21. “Coming_soon” Relation:

- **Attributes:**

Coming_soon {Content_ID}

Here **Primary key** = { Content_ID }

According to theorem, all attribute primary key relation is always in BCNF.

So “Coming_soon” is in BCNF.